

Rayan Alam

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EDUCATION

University of California, Los Angeles

M.S in Mechanical Engineering

Expected Dec 2026

Arizona State University

B.S. in Biomedical Engineering – *Cum Laude*

May 2025

Dean's List: Fall 2021- Fall 2022, Spring 2024-Spring 2025

SKILLS

- **Testing & Validation:** system and component-level **validation test plans**, Instron compression testing, durability and pressure-cycle testing, comparative validation testing
- **Root Cause Analysis:** iterative process characterization, **root cause analysis**, comparative parameter and material testing to isolate failure modes
- **Electromechanical Systems:** custom PCB design for safety-critical control systems, **control system integration**
- **Data Analysis & Documentation:** MATLAB, statistical analysis, **test data analysis**, technical reports
- **CAD & Prototyping:** SolidWorks, Fusion 360, AutoCAD, 3D printing, silicone molding

EXPERIENCE

Bionics Lab, UCLA- Research Assistant

September 2025 - Present

- Participate in research on medical and wearable robotics, with emphasis on biomechanics, control systems, and human-machine interaction.
- Designed a custom PCB to regulate a power/safety cutoff switch for the control system, enabling operators to immediately de-energize the teleoperation rig during testing and improving overall system safety.
- Support **3** human-subject studies on VR-based rehabilitation by managing experimental setup and data collection, maintaining **95%** protocol compliance across sessions.
- Assist in experimental setup, data collection, and system integration for human-subject studies.

Weaver Lab: Biomaterials and Immune Engineering-Research Assistant

August 2023 - May 2025

- Designed and iterated device molds across **15** 3D-printed prototypes using **4** different print materials, identifying the optimal combination for dimensional accuracy through comparative testing.
- Performed flow simulations in SolidWorks to evaluate mass transport, informing geometry optimization that improved oxygen diffusion uniformity by **18%**.
- Achieved a consistent **50/50** material split for a Type 1 diabetes islet-encapsulation device by identifying **0.5 mL/min** as the optimal pump speed across dual-syringe co-extrusion of alginate and gluconic-delta-lactone (GDL) hydrogel trials (tested at **0.5, 1, and 3 mL/min**).

Massachusetts General Hospital (MGH) - Harvard Medical School-Research Assistant

June 2024 - August 2024

- Supported translational research on enteric neural stem cell (ENSC) transplantation in a mouse model of Hirschsprung disease.
- Assisted in isolating, expanding, and microinjecting enteric neural stem cells (ENSCs) as neurospheres into the aganglionic colon of Ednrb knockout mice, achieving cell coverage of up to **16.8 mm²** following multiple-injection delivery via laparotomy.
- Conducted literature reviews and contributed to histological and motility analysis, showing that ENSC transplantation restored colonic motor complex frequency and extended median survival from **10 to 26 days**

PROJECTS & RESEARCH

Implantable Leptin Delivery Pump-Prototype Lead

August 2024 - May 2025

- Led development of an implantable pump delivering leptin into cerebrospinal fluid to prevent Type II diabetes, engineering a **0.3 mL/day** dosing profile that held within a **0.2–0.4 mL/day** acceptance range across three physiological temperatures (**97–102°F**) over **4-day trials**, repeated across **3 independent runs**.
- Designed a full parametric CAD assembly in SolidWorks and selected device materials (Platinum Cure Silicone, ABS, SIL-Poxy), including a custom silicone bellow for controlled, low-flow drug delivery; applied design-for-manufacturability (DFM) practices to address low-volume silicone molding and tooling constraints.
- Validated durability via Instron compression testing across **10–500 kPa (15 total pressure cycles)** as part of a **verification and validation (V&V) protocol**, achieving structural integrity scores of **9–10** out of **10** with zero leaks or failures — exceeding the project's $\geq 8/10$ reusability threshold. .

CleanWater Toilets-Prototype Designer

August 2021 -December 2021

- Cut assembly time **40%** for non-technical stakeholders by redesigning a bag-and-frame toilet system into a single-step snap-fit Fusion 360 CAD assembly.
- Designed a low-cost, bag-based dispensable toilet system addressing open defecation for an estimated **154 million** people in rural India, and pitched a full business plan sizing the addressable market at **34 million** underserved rural households with **5%** projected penetration over **3 years**.